

## ABBOTTABAD KAKUL, Pakistan

WMO: 415350

Lat: 34.183N

Lon: 73.250E

Elev: 1309

StdP: 86.56

Time Zone: 5.00 (E05)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.6       | 0.4 | -6.4                          | 2.6 | 7.2  | -5.1 | 2.9 | 7.0  | 2.0                   | 12.0 | 1.5 | 11.7 | 0.0                   | N/A  | 0.315 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 13.0                   | 34.1            | 20.3 | 32.7 | 20.5 | 31.3 | 20.8 | 24.6                | 29.3 | 24.0 | 28.7 | 23.5 | 28.0 | 0.7                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.3                            | 21.3 | 27.6 | 22.8 | 20.6 | 27.0 | 22.2 | 19.8 | 26.4 | 83.4          | 29.8 | 80.6 | 28.5 | 78.2 | 28.6 | 28.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 1.8           | 1.5  | 1.3 | -2.0                       | 36.6 | 0.8                | 1.2 | -2.6                                               | 37.5 | -3.1       | 38.2 | -3.5       | 38.8 | -4.1       | 39.7 |    |    |
| (5)               |      |     | -3.6                       | 25.9 | 0.9                | 0.8 | -4.3                                               | 26.5 | -4.8       | 27.0 | -5.4       | 27.5 | -6.0       | 28.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.0   | 7.1   | 8.5   | 12.8  | 17.5  | 21.9  | 24.8  | 24.9  | 24.1  | 22.2  | 18.1  | 12.8  | 8.9   |  |  |
|                                                                        | DBStd       | 6.91   | 2.35  | 2.71  | 3.59  | 3.29  | 2.65  | 2.30  | 1.67  | 1.58  | 1.73  | 2.69  | 2.29  | 2.17  |  |  |
|                                                                        | HDD10.0     | 218    | 94    | 56    | 14    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 48    |  |  |
|                                                                        | HDD18.3     | 1356   | 349   | 276   | 176   | 53    | 4     | 0     | 0     | 0     | 1     | 38    | 167   | 293   |  |  |
|                                                                        | CDD10.0     | 2772   | 4     | 14    | 99    | 226   | 369   | 445   | 462   | 437   | 365   | 250   | 88    | 13    |  |  |
|                                                                        | CDD18.3     | 869    | 0     | 0     | 3     | 28    | 115   | 195   | 203   | 179   | 116   | 29    | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 7932   | 0     | 0     | 43    | 342   | 1242  | 2147  | 1740  | 1306  | 827   | 280   | 6     | 0     |  |  |
| (13)                                                                   | CDH26.7     | 2786   | 0     | 0     | 2     | 81    | 492   | 1022  | 621   | 350   | 179   | 38    | 0     | 0     |  |  |
| (14)                                                                   | Wind        | WSAvg  | 0.3   | 0.2   | 0.3   | 0.3   | 0.3   | 0.4   | 0.3   | 0.3   | 0.3   | 0.2   | 0.2   | 0.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1331   | 71    | 115   | 125   | 113   | 77    | 96    | 249   | 257   | 100   | 52    | 31    | 46    |  |  |
|                                                                        | PrecMax     | 1615   | 171   | 217   | 200   | 254   | 165   | 375   | 495   | 373   | 205   | 173   | 95    | 129   |  |  |
|                                                                        | PrecMin     | 1030   | 0     | 7     | 16    | 11    | 28    | 10    | 85    | 130   | 5     | 1     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 179    | 49    | 57    | 52    | 67    | 37    | 93    | 88    | 67    | 50    | 48    | 32    | 34    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 18.2  | 21.1  | 26.6  | 31.4  | 34.7  | 36.9  | 34.8  | 31.7  | 30.6  | 29.3  | 24.1  | 20.1  |  |  |
|                                                                        |             | MCWB   | 8.3   | 10.3  | 15.2  | 18.0  | 19.0  | 20.6  | 21.7  | 24.0  | 21.4  | 19.0  | 12.8  | 9.7   |  |  |
|                                                                        | 2%          | DB     | 16.4  | 18.3  | 24.5  | 29.0  | 32.9  | 34.9  | 32.9  | 30.5  | 29.4  | 27.8  | 22.1  | 18.1  |  |  |
|                                                                        |             | MCWB   | 7.5   | 9.1   | 14.3  | 16.9  | 18.7  | 20.6  | 22.4  | 23.5  | 21.1  | 17.8  | 12.2  | 9.0   |  |  |
|                                                                        | 5%          | DB     | 14.9  | 16.4  | 22.6  | 26.9  | 31.2  | 33.4  | 31.4  | 29.7  | 28.6  | 26.3  | 20.6  | 16.5  |  |  |
|                                                                        |             | MCWB   | 6.9   | 8.5   | 13.0  | 16.3  | 18.4  | 20.4  | 22.5  | 23.3  | 20.8  | 17.2  | 11.8  | 8.1   |  |  |
|                                                                        | 10%         | DB     | 13.1  | 14.6  | 20.4  | 25.1  | 29.7  | 32.0  | 30.0  | 28.8  | 27.8  | 24.8  | 19.0  | 14.8  |  |  |
|                                                                        |             | MCWB   | 6.1   | 7.9   | 12.2  | 15.5  | 17.9  | 20.3  | 22.9  | 23.0  | 20.6  | 16.0  | 10.8  | 7.4   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 9.4   | 11.7  | 16.5  | 19.0  | 20.9  | 24.6  | 25.2  | 25.5  | 23.8  | 21.4  | 14.6  | 10.6  |  |  |
|                                                                        |             | MCDB   | 14.4  | 17.6  | 23.5  | 29.6  | 30.5  | 31.0  | 30.2  | 30.1  | 28.5  | 27.0  | 20.7  | 17.4  |  |  |
|                                                                        | 2%          | WB     | 8.4   | 10.4  | 15.0  | 17.6  | 19.8  | 23.7  | 24.5  | 24.7  | 22.7  | 20.0  | 13.5  | 9.6   |  |  |
|                                                                        |             | MCDB   | 14.2  | 16.1  | 22.9  | 27.1  | 30.3  | 29.9  | 29.4  | 29.0  | 27.7  | 25.9  | 20.3  | 16.1  |  |  |
|                                                                        | 5%          | WB     | 7.5   | 9.3   | 13.7  | 16.9  | 19.0  | 22.8  | 24.0  | 24.1  | 22.1  | 18.3  | 12.5  | 8.7   |  |  |
|                                                                        |             | MCDB   | 13.3  | 14.8  | 21.3  | 25.5  | 29.7  | 29.2  | 28.7  | 28.4  | 27.2  | 24.1  | 19.1  | 15.2  |  |  |
|                                                                        | 10%         | WB     | 6.7   | 8.4   | 12.6  | 16.1  | 18.2  | 22.0  | 23.5  | 23.5  | 21.4  | 16.9  | 11.4  | 7.9   |  |  |
|                                                                        |             | MCDB   | 12.0  | 13.2  | 19.3  | 24.0  | 28.3  | 28.5  | 27.9  | 27.5  | 26.1  | 23.1  | 17.5  | 14.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.4  | 10.7  | 11.8  | 12.8  | 13.6  | 13.0  | 9.4   | 8.9   | 10.9  | 13.5  | 12.9  | 12.4  |  |  |
|                                                                        |             | MCDBR  | 14.5  | 14.4  | 15.2  | 15.7  | 16.0  | 15.4  | 12.2  | 10.4  | 12.4  | 14.6  | 15.0  | 15.0  |  |  |
|                                                                        |             | MCWBR  | 7.5   | 7.0   | 7.0   | 6.9   | 6.1   | 6.0   | 5.3   | 5.0   | 5.8   | 6.2   | 7.0   | 7.3   |  |  |
|                                                                        |             | MCDBR  | 12.5  | 12.2  | 14.2  | 14.5  | 14.8  | 11.4  | 9.3   | 8.8   | 10.2  | 12.7  | 13.2  | 13.4  |  |  |
|                                                                        | 5% WB       | MCWBR  | 7.0   | 6.6   | 7.4   | 6.9   | 6.2   | 5.5   | 4.9   | 4.7   | 5.5   | 6.4   | 6.9   | 7.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.371 | 0.420 | 0.431 | 0.444 | 0.460 | 0.524 | 0.588 | 0.563 | 0.475 | 0.426 | 0.398 | 0.366 |  |  |
|                                                                        | taud        |        | 2.129 | 2.029 | 2.054 | 2.072 | 2.050 | 1.912 | 1.816 | 1.875 | 2.047 | 2.078 | 2.084 | 2.161 |  |  |
|                                                                        | Ebn at Noon |        | 809   | 807   | 837   | 846   | 834   | 778   | 726   | 735   | 786   | 790   | 769   | 789   |  |  |
|                                                                        | Edh at Noon |        | 121   | 150   | 159   | 163   | 169   | 194   | 212   | 197   | 158   | 141   | 126   | 111   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.79  | 3.28  | 4.53  | 5.70  | 6.86  | 7.03  | 6.19  | 5.62  | 5.43  | 4.68  | 3.40  | 2.74  |  |  |
|                                                                        | RadStd      |        | 0.35  | 0.53  | 0.50  | 0.40  | 0.44  | 0.36  | 0.47  | 0.44  | 0.34  | 0.29  | 0.26  | 0.26  |  |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (6 neighbors) | +0.62 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page